AIML-401: Deep Learning — Revision Study Guide

Topic: **Backpropagation Algorithm & Gradient Flow**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 1: Mathematical Foundations	**Compiled:** Nov 02, 2026 - 18:59
Faculty Note Ref: syllabus_overview.pdf	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Definition

Backpropagation is an exact algorithmic implementation of reverse-mode automatic differentiation applied to composite loss functions in multi-layer perceptrons (MLP). Its purpose is to evaluate the exact gradient of the scalar error L with respect to every weight parameter in O(N) operations per pass.

2. Mathematical Formulation & Chain Rule Derivation

For an output node k with net excitation $z_k = \sum w_{ik} a_i + b_k$ and activation $y_k = \sigma(z_k)$, the error gradient flows recursively:

$$\partial L / \partial w_{jk} = (\partial L / \partial y_k) \cdot \sigma'(z_k) \cdot a_j = \delta_k \cdot a_j$$
$$\delta_j = (\sum \delta_k w_{jk}) \cdot \sigma'(z_j)$$

Key Exam Insight: *The error term δ at hidden layer j equals the weighted sum of downstream errors propagated back across connecting forward weights, scaled by the activation derivative $\sigma'(z_j)$.*

3. 4-Phase Algorithmic Execution Pipeline

Phase	Name	Mathematical Operation	Complexity
1	Forward Pass	Compute layer inputs z = W a + b and activations y = f(z)	O(W) Matrix Multiply
2	Loss Evaluation	Compute error: L = (1/2) * sum((y_pred - y_true)^2)	O(K) Vector Norm
3	Backward Pass	Evaluate output delta: delta_k = (y_pred - y_true) * f'(z_k), propagate back	O(W) Transposed Multiply
4	Weight Update	W_new = W_old - eta * (delta * a_prev^T) via Gradient Descent	O(W) Elementwise Add

4. Faculty Grounded Citations (Direct from Professor Uploads)

Source Document: *API Test Syllabus Overview (Uploaded by faculty FAC_AIML_01)*
“Unit 1 multivariable calculus and matrix operations foundations....”

5. High-Yield Question & Pitfall Checklist for IA-2 / SEE

Top Exam Trap: Forgetting to include the activation derivative σ'(z) in the chain rule expansion.
Expected Question (8 Marks): *'Derive the backpropagation weight update equation for a hidden layer in a 3-layer MLP using sigmoid activations and MSE loss.'*
Marking Rubric:
• 2 Marks: Clear signal flow and error backpropagation diagram.
• 3 Marks: Full expansion of $\partial L/\partial w = \partial L/\partial z \cdot \partial z/\partial w$.
• 3 Marks: Vectorized final update rule $W \leftarrow W - \eta\, \delta\, (a)^T$.